

Complete AI Agent Evaluation Framework

AWS Bedrock AgentCore · Strands Evals · Arize Phoenix

EU Banking Edition — RAI · PII · GDPR · DORA · EBA · AMLD6

Ideation to Retirement: The Definitive Production Guide

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Document Purpose — This is the definitive production guide for evaluating AI agents built on AWS Bedrock AgentCore and the Strands Agents SDK, instrumented via Arize Phoenix, and deployed in EU banking environments subject to the EU AI Act, GDPR, DORA, and EBA guidelines. Every evaluation dimension is covered: quality, safety, RAI, PII, fairness, explainability, drift detection, and automated pipelines — from first idea to system retirement.

10 Strands Evals Evaluators	13 AgentCore Evaluators	4 Eval Levels	9 Lifecycle Phases	3 Regulatory Regimes	EUR 35M Max AI Act Penalty
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1.1 The Evaluation Problem in AI Agents

Traditional software testing assumes deterministic outputs: same input, same expected output. AI agents break this completely — they generate natural language, make context-dependent decisions, call tools in adaptive sequences, and produce varied outputs from identical inputs. Evaluation therefore requires **judgment** rather than keyword comparison and **multi-dimensional scoring** rather than binary pass/fail.

WHY AGENTS ARE HARD TO EVALUATE

- Natural language has many valid forms — no single correct answer exists
- Agents take multi-step actions; final response alone misses intermediate failures
- Multi-turn conversations need session-level context awareness
- Tool selection and parameter correctness add agentic dimensions absent in LLM evals
- Non-determinism means the same test case can produce different results each run

1.2 Component Map — Three Primary Layers

The framework integrates components across three layers: native SDK evaluation (Strands Evals), managed AWS evaluation service (AgentCore Evaluations), and open-source observability (Arize Phoenix), plus supporting specialist libraries.

Layer	Component	Primary Role	When to Use
Native SDK	Strands Evals	Cases, Experiments, 10 evaluators, ActorSimulator, CI/CD gates	Development, CI/CD, regression testing
Managed AWS	AgentCore Evaluations	13 built-in + custom evaluators; online (production) + on-demand modes	Pre-deployment gates & continuous production scoring
Observability	Arize Phoenix	OTEL traces, embedding drift, human annotation, experiment tracking	Debug, drift detection, production visibility
RAG Eval	RAGAS	Faithfulness, context precision/recall, answer relevancy	Any RAG pipeline quality assessment
Unit Testing	DeepEval	Pytest-style LLM tests, DAG metric for agentic planning	CI/CD quality gates, regression suite
Trace & Cost	Langfuse	Strands-native traces, cost per session, prompt management	Development observability, cost optimisation

1.3 Decision: Which Tool for Which Job

Scenario	Primary Tool	Supporting Tool
Pre-deployment quality gate	Strands Evals (CI/CD)	AgentCore On-Demand
Production sampling & scoring	AgentCore Online Eval	Phoenix Online Evals
RAG pipeline quality	RAGAS	Phoenix RAG templates
Multi-turn conversation testing	Strands Evals ActorSimulator	DeepEval DAG metric
Debug a specific user complaint	Phoenix Trace Explorer	AgentCore On-Demand
Detect gradual performance decay	Phoenix Drift Monitors	CloudWatch EWMA alarms
EU compliance gate (PII, fairness)	Custom Strands Evaluators	AgentCore Custom Evaluator ARN
Human review & annotation	Phoenix Annotation Queue	Argilla
A/B prompt comparison	Phoenix Experiments	Langfuse + RAGAS

Scenario	Primary Tool	Supporting Tool
AML / SAR quality check	Custom Strands Evaluator	AgentCore Custom Evaluator
PII leakage detection	AgentCore Builtin.PIILeakage	Presidio + AWS Comprehend (pre-filter)
Fairness / bias audit	Custom Counterfactual Cases	DeepEval Bias Metric

CHAPTER 2 — STRANDS EVALS: NATIVE SDK REFERENCE

2.1 Core Abstractions

Strands Evals (github.com/strands-agents/evals) is the first-party evaluation framework built specifically for the Strands Agents SDK. It introduces three foundational abstractions plus the Task Function bridge pattern (AWS ML Blog, March 18 2026).

Concept	Definition	Analogy
Case	Single test scenario: input + optional expected_output + expected_trajectory + metadata	Unit test case
Experiment	Bundle of Cases + Evaluators that orchestrates execution and generates EvaluationReports	Test suite / pytest session
Evaluator	LLM-based or code-based judge that scores output against rubric or expected value	Test assertion with judgment
Task Function	Callable you provide to Experiment — invokes agent live (online) or retrieves saved traces (offline)	Test fixture / mock setup
ActorSimulator	LLM-powered simulated user that drives realistic multi-turn conversations to a defined goal	Synthetic user in usability testing
ExperimentGenerator or	LLM that generates diverse Cases and rubrics from a high-level context description	Automated test case generator
EvaluationReport	Per-case and per-experiment score breakdowns, pass/fail status, evaluator reasoning	JUnit XML / pytest report

2.2 All 10 Built-in Evaluators

Evaluator	Family	Level	Scale	Description	EU Banking Use
OutputEvaluator	Rubric	Trace	0.0-1.0	Judges final response via custom natural-language rubric	GDPR Art.22 adverse action notice quality; AI disclosure check
TrajectoryEvaluator	Rubric	Trace	0.0-1.0	Judges tool-call sequence via 3 scoring modes: exact / in-order / any-order match	AML: PII-check -> AML-score -> HITL (exact match required by AMLD6)
InteractionsEvaluator	Rubric	Session	0.0-1.0	Evaluates sequences of interactions between multiple agents or components	KYC agent -> credit agent handoff audit trail (EU AI Act Art.12)
HelpfulnessEvaluator	Semantic	Trace	1-7 scale	Pre-built 7-point scale: user value actually delivered, not just technical correctness	EBA consumer outcomes — genuine utility measure
FaithfulnessEvaluator	Semantic	Trace	1-5 scale	Pre-built 5-point: response grounded in conversation history / context (anti-hallucination)	GDPR accuracy — no hallucinated financial or personal data
HarmfulnessEvaluator	Semantic	Trace	Binary Y/N	Pre-built binary: harmful / inappropriate / dangerous content check	EU AI Act prohibited practices gate — always 100% sampling
ToolSelectionAccuracyEvaluator	Tool	Tool	0.0-1.0	Was the correct tool chosen for the task at this point in conversation?	CRR/CRD: credit_model tool must be called for credit decisions
ToolParameterAccuracyEvaluator	Tool	Tool	0.0-1.0	Were parameters passed to each tool correct and complete?	Wrong customer_id -> wrong credit data -> GDPR Art.22 violation risk
GoalSuccessRateEvaluator	Session	Session	0.0-1.0	Broadest view: did the user ultimately achieve their stated goal?	EBA consumer outcome: did customer resolve their issue end-to-end?

Evaluator	Family	Level	Scale	Description	EU Banking Use
Custom (via OutputEvaluator)	Domain	Any	Define own	Any domain-specific rubric via OutputEvaluator or TrajectoryEvaluator	GDPR Art.22, AMLD6 SAR, fairness/DIR, PII leakage, AI disclosure

2.3 Online vs Offline Task Patterns

Pattern	How It Works	Best For	EU Banking Note
Online (live invocation)	Task function creates agent, runs case live, captures response and execution trace for scoring	CI/CD gates, active development, regression checks before deployment	Run counterfactual fairness suite before every deployment merge
Offline (historical traces)	Task function retrieves recorded traces from Phoenix / Langfuse / DB; parses into evaluator format	Production traffic analysis; A/B model comparison on real interactions	DORA audit: evaluate 6-month trace archive for historical compliance
Async parallel	experiment.run_evaluations_async(task_fn, max_workers=N) enables concurrent evaluation at scale	Large benchmark suites (200-1000 cases) in time-boxed CI/CD pipeline	Run 500-case EU compliance suite in under 10 minutes with max_workers=20

2.4 ActorSimulator — Multi-Turn Testing

Real users do not follow scripts. ActorSimulator generates a realistic user profile — personality, expertise level, communication style, and a goal — then drives the conversation automatically. It emits a stop token when the goal is achieved, or stops at max_turns. The transcript is passed to GoalSuccessRateEvaluator to assess end-to-end success.

ACTORSIMULATOR — EU BANKING APPLICATIONS

- Simulate confused retail customer attempting loan application across 8 turns
- Simulate sophisticated customer testing AML flag explanation (tests Art.13 transparency)
- Simulate regulatory examiner probing credit decision explanation (tests GDPR Art.22)
- Simulate adversarial user attempting prompt injection (tests EU AI Act Art.15 robustness)
- Simulate customer exercising GDPR Art.17 right-to-erasure request via chat
- Simulate non-native speaker with language difficulty (tests accessibility and fairness)

2.5 TrajectoryEvaluator — Three Scoring Modes

The TrajectoryEvaluator provides an LLM judge with three built-in scoring functions. The judge selects the most appropriate mode based on the rubric provided. This is critical for EU banking where regulatory compliance requires specific tool ordering.

Scoring Mode	Definition	When to Use	EU Banking Example
Exact match	Actual tool sequence must match expected sequence precisely — same tools, same order, nothing extra	When tool order is critical to safety, compliance, or legal obligation	AML: pii_check THEN aml_score THEN human_escalate — no exceptions
In-order match	Expected tools appear in order but other tools may be interspersed between them	When required steps must happen in sequence but other calls are acceptable	Credit: credit_check must precede decision, but explanation calls may occur anywhere
Any-order match	All expected tools appear somewhere in the trajectory — order is flexible	When all required checks must occur but sequence is not legally prescribed	Eligibility check: income_check, employment_check, id_verify — any order fine

CHAPTER 3 — AGENTCORE EVALUATIONS: MANAGED SERVICE REFERENCE

Amazon Bedrock AgentCore Evaluations is a **fully managed, continuous assessment service**. It operates via two modes — online (live traffic) and on-demand (targeted) — and integrates with any agent framework via OpenTelemetry. Results publish to Amazon CloudWatch in real time, enabling alarms, automations, and dashboards.

3.1 Online Evaluation — Continuous Production Monitoring

Online evaluation samples live production traffic and scores interactions using LLM-as-a-Judge. It provides persistent monitoring across multiple quality dimensions simultaneously.

Configuration Parameter	Options	EU Banking Recommendation
Sampling rate	Percentage-based (e.g. 10%) or conditional filters on session metadata	Start 10%; auto-increase to 50% on drift alert; always 100% for PII and harmfulness evaluators
Evaluator selection	Built-in evaluator ARNs or custom evaluator ARNs	Helpfulness + ToolSelectionAccuracy + Groundedness + Custom PIILeakage + Custom Fairness
Session filtering	Include/exclude by session metadata, user type, agent version tag	Exclude test sessions; include only production customer-facing sessions via metadata filter
Output destination	Amazon CloudWatch metrics (default); S3 for raw score archive	AgentCore Observability Dashboard + CloudWatch Alarms + Lambda auto-remediation
IAM role	Create new AWS IAM role or pass existing one with least-privilege policy	Separate IAM role per agent for auditability; log all role assumption events in CloudTrail

3.2 On-Demand Evaluation — Targeted Assessment

On-demand evaluation assesses specific interactions by analysing a chosen set of span or trace IDs. Unlike online evaluation, it is triggered explicitly and processes only the interactions you specify.

ON-DEMAND EVALUATION USE CASES

- CI/CD pipeline: evaluate golden dataset before every deployment merge
- Incident investigation: evaluate specific customer interactions flagged by a complaint
- A/B comparison: same trace IDs evaluated with two different evaluator rubrics
- Regulatory audit: evaluate 90-day sample of AML decision traces on demand
- Custom evaluator testing: validate a new rubric before enabling it in online mode
- Historical compliance: evaluate archived traces against new EU AI Act requirements

3.3 All 13 Built-in AgentCore Evaluators

Evaluator ARN Suffix	Dimension	Scale	Description
Builtin.Helpfulness	Quality	0-1	Does the response deliver genuine user value? User-perspective quality assessment.
Builtin.Correctness	Quality	0-1	Factual accuracy compared to ground truth or retrieved context.
Builtin.Coherence	Quality	0-1	Logical flow, internal consistency, and readability of the response.
Builtin.Completeness	Quality	0-1	Does the response address all aspects and sub-questions in the user query?
Builtin.Groundedness	Quality	0-1	All claims in the response are supported by retrieved context. Anti-hallucination metric.
Builtin.Relevance	Quality	0-1	Response is relevant and on-topic relative to the user's actual question.
Builtin.ToolSelectionAccuracy	Tool	0-1	Was the correct tool chosen for the task at each step of the conversation?

Evaluator ARN Suffix	Dimension	Scale	Description
Builtin.ToolParameterAccuracy	Tool	0-1	Were correct and complete parameters passed to each tool invocation?
Builtin.ToolCallSequence	Tool	0-1	Tools invoked in the correct logical and/or legally required order?
Builtin.GoalSuccessRate	Session	0-1	Did the user ultimately achieve their stated goal? Session-level success.
Builtin.Harmlessness	Safety	0-1	Response is free from harmful, abusive, violent, or dangerous content.
Builtin.RefusalQuality	Safety	0-1	When agent declines, does it help user find an appropriate alternative?
Builtin.PIILeakage	Safety	0-1	No personally identifiable information exposed in the output. Zero tolerance.

3.4 Custom Evaluators

Custom evaluators extend AgentCore's evaluation capability with domain-specific quality metrics. Each has a unique ARN and can be referenced in both online and on-demand evaluation configurations.

Parameter	Description	EU Banking Example
model_id	Bedrock model ARN for the LLM judge	anthropic.claude-sonnet-4-20250514-v1:0 with temperature=0.0 for deterministic scoring
evaluation_prompt	Full system prompt: rubric, scoring scale, output schema (must be JSON)	GDPR Art.22 adverse action quality rubric with 1.0/0.5/0.0 scoring scale and critical_fail conditions
scoring_schema	Type (numeric/binary/categorical), pass_threshold, composite weight map	{score: 0-1, critical_issues: [], reasoning: ""} with pass_threshold: 0.75
critical_fail_on	Conditions that hard-fail the evaluation regardless of overall score	compliance_score < 0.5 OR pii_detected = true — either triggers BLOCK and DPO notification

3.5 CloudWatch Dashboard Specification

Dashboard Row	Widgets	Alert Condition
Quality Scores (7d trend)	Helpfulness, Correctness, Groundedness, Completeness — line chart per evaluator	Any metric below threshold for 3 consecutive evaluation periods
Tool Accuracy (7d)	ToolSelection, ToolParameter, ToolCallSequence — line chart	Any tool metric below 0.85 for 2 periods
Safety Gates (24h)	Harmlessness, PIILeakage, RefusalQuality — bar chart by hour	ANY single safety failure = CRITICAL alert — immediate DPO notification
Operational Metrics	Latency P50/P95/P99, token count, error rate — multi-line	P95 latency > 3000ms or error rate > 1%
Goal Success Rate	GoalSuccessRate trend, session duration, active session count	GSR below 0.70 for 2 consecutive periods
Cost Tracking	Cost per session (estimated from token usage) — area chart	Cost exceeds 1.5x rolling 7-day average
Drift Indicators	Sampling rate (auto-adjusted), eval coverage %, embedding drift score	Coverage drop below 5% or embedding drift score exceeds 0.15

CHAPTER 4 — ARIZE PHOENIX: OBSERVABILITY & EVALUATION LAYER

Arize Phoenix is an open-source, vendor-agnostic AI observability platform built on OpenTelemetry. For EU banking it is self-hostable on AWS EKS in Frankfurt (eu-central-1) to satisfy GDPR data residency. It provides end-to-end traces, embedding drift detection, LLM-as-a-Judge evaluations, human annotation queues, and experiment management.

4.1 OpenTelemetry Span Types

Span Type	What It Captures	Evaluation Dimension
LLM	Model ID, prompt/completion tokens, latency, input messages, output text, estimated cost	Faithfulness, Helpfulness, Harmlessness, Correctness
Retriever	Query text, retrieved documents, relevance scores, document metadata	Context Precision, Context Recall, Context Relevance
Tool	Tool name, input parameters, output, execution time, error messages	ToolSelection, ToolParameter, ToolCallSequence
Chain	Sequence of steps, cumulative token count, total latency, intermediate outputs	Trajectory quality, Step efficiency ratio
Agent	Full session, all sub-spans, final response, goal metadata, user feedback	GoalSuccessRate, Conversation quality
Embedding	Input text, embedding vector, model ID, vector dimensions	Embedding drift detection — semantic shift monitoring

4.2 Drift Detection System

Drift Type	What Shifts	Detection Method	Phoenix Tool	Alert Threshold
Input embedding drift	User query distribution changes semantically (new topics, new user segments)	Cosine distance from 7-day baseline centroid, computed hourly	InputEmbeddingDriftMonitor	> 0.15 drift score
Output embedding drift	Response semantic distribution shifts (tone, scope, vocabulary changes)	Euclidean distance in embedding space from rolling baseline	OutputEmbeddingDriftMonitor	> 0.20 drift score
Performance drift	Evaluation scores decline gradually over days or weeks	EWMA control chart ($\lambda=0.1$); UCL/LCL at 3 sigma	MetricDriftMonitor	UCL or LCL breach
Step-change drift	Sudden quality drop from model update or prompt change	CUSUM control chart ($k=0.5$ sigma, $h=5$ sigma)	MetricDriftMonitor	CUSUM stat > threshold
Concept drift	Knowledge base becomes stale relative to real-world facts	Faithfulness decline + RAG context recall drop combined signal	Combined monitor	Faithfulness < 0.75 sustained 3d
Data drift	Feature distribution of structured inputs shifts	KL divergence and PSI on input features	FeatureDriftMonitor	PSI > 0.20

CHAPTER 5 — COMPLETE METRIC CATALOGUE

5.1 Response Quality Metrics

Metric	Primary Tool	Scale	Target	Alert
Helpfulness	AgentCore Builtin / Strands Evals	0-1	≥ 0.80	< 0.75
Correctness	AgentCore Builtin / DeepEval GEval	0-1	≥ 0.82	< 0.70
Coherence	AgentCore Builtin	0-1	≥ 0.82	< 0.72
Completeness	AgentCore Builtin	0-1	≥ 0.80	< 0.70
Groundedness	AgentCore Builtin	0-1	≥ 0.85	< 0.75
Answer Relevancy	RAGAS	0-1	≥ 0.82	< 0.72
Hallucination Rate	Phoenix / DeepEval	0-1	≤ 0.10	> 0.20
Toxicity	DeepEval / Phoenix	0-1	≤ 0.01	> 0.05
Bias Score	DeepEval / Custom	0-1	≤ 0.05	> 0.10
Conciseness	DeepEval GEval	0-1	≥ 0.75	< 0.60

5.2 RAG Quality Metrics (RAGAS)

Metric	Method	Target	Alert	What It Catches
Faithfulness	Claims_supported / Total_claims	≥ 0.85	< 0.75	Hallucination — claims not in context
Context Precision	Relevant_chunks / Retrieved_chunks	≥ 0.80	< 0.70	Retriever noise — irrelevant chunks
Context Recall	GT_in_context / GT_total	≥ 0.80	< 0.70	Missing evidence — incomplete retrieval
Context Relevance	LLM-Judge scoring	≥ 0.80	< 0.70	Off-topic retrieved chunks
Answer Relevancy	Cosine similarity: question vs answer	≥ 0.82	< 0.72	Response misaligned with query intent
Response Groundedness	LLM-Judge per-claim check	≥ 0.85	< 0.75	Claims lacking context support
Noise Sensitivity	Score(noisy) minus Score(clean)	≤ 0.10	> 0.20	Fragile retrieval affected by noise
Agent Goal Accuracy	LLM-Judge: goal vs achieved outcome	≥ 0.80	< 0.65	Agent fails to meet stated user goal
Topic Adherence	LLM-Judge: on-scope responses ratio	≥ 0.90	< 0.80	Agent answers out-of-scope questions

5.3 Agentic Behaviour Metrics

Metric	Primary Tool	Target	Alert	Description
Tool Selection Accuracy	AgentCore / Strands Evals	≥ 0.90	< 0.80	Correct tool chosen for each step
Tool Parameter Accuracy	AgentCore / Strands Evals	≥ 0.88	< 0.78	Correct parameters extracted from query
Tool Call Sequence	AgentCore / TrajectoryEvaluator	≥ 0.90	< 0.80	Tools called in correct legal order
Task Success Rate	Custom / DeepEval DAG	≥ 0.75	< 0.60	Binary: user goal achieved end-to-end
Planning Quality	DeepEval DAG metric	≥ 0.80	< 0.65	Correct goal decomposition into steps
Step Efficiency	Custom: actual / min steps	≥ 0.80	< 0.65	Not wasting unnecessary tool calls

Metric	Primary Tool	Target	Alert	Description
Goal Success Rate	AgentCore / Strands Evals	>= 0.75	< 0.60	Session-level user goal achievement
Conversation Quality	RAGAS	>= 0.80	< 0.68	Multi-turn coherence and memory use

5.4 Operational & Infrastructure Metrics

Metric	Source	Target	Alert	DORA Requirement
Agent Uptime	CloudWatch	>= 99.5%	< 99.0%	DORA ICT availability obligation
Latency P50 (ms)	CloudWatch	<= 1200	> 2000	DORA performance resilience
Latency P95 (ms)	CloudWatch	<= 2500	> 3500	DORA performance resilience
Error Rate (%)	CloudWatch	<= 0.5%	> 1.0%	DORA incident classification threshold
MTTD (min)	CloudWatch	<= 15	> 30	DORA: detect incidents within SLA
MTTR (hours)	CloudWatch	<= 4	> 8	DORA: respond and restore within SLA
Token cost / session (USD)	CloudWatch	<= 0.05	> 0.10	Budget governance (not DORA)
Tool failure rate (%)	Phoenix	<= 2%	> 5%	Downstream API reliability (DORA Art.28)
Eval coverage (%)	AgentCore	>= 10%	< 5%	Monitoring adequacy for ongoing oversight

CHAPTER 6 — BENCHMARK CREATION STANDARDS

6.1 Three-Tier Benchmark Architecture

Tier	Type	Examples	Purpose	Update Frequency
Tier 1	Public benchmarks	MMLU, HumanEval, SWE-bench, BrowserBench, GAIA, TruthfulQA, HellaSwag, FinanceBench	Calibrate against industry baselines; compare model versions objectively	At every model upgrade
Tier 2	Domain benchmarks	FinanceBench, MedQA, LegalBench, custom BankingBench built from EU regulatory scenarios	Vertical-specific calibration for banking use cases	Quarterly
Tier 3	Custom / golden dataset	Curated from real production traffic (anonymised) + SME-annotated + adversarial red-team cases	Production truth — the definitive gate that blocks deployment	Monthly + on every incident

6.2 Golden Dataset Construction

Step	Actions	Quality Gate
A: Seed Collection	Real user queries from Phoenix (anonymise PII first); SME-authored exemplar questions; synthetic cases from ExperimentGenerator; adversarial red-team cases; existing regression test cases	Target 100-1000 examples per agent/domain; minimum 20% adversarial cases
B: Ground Truth Annotation	Expected answer per case; acceptable variant answers; quality rubric 1-5 scale; metadata: difficulty, domain, query type (RAG/tool/reasoning/safety)	Inter-annotator Cohen's Kappa ≥ 0.70 required; adjudicate disagreements via senior SME
C: Stratified Split	Train Eval 70% (prompt optimisation); Dev Eval 15% (model selection, never shown to evaluators); Test/Golden 15% (final gate only, never used during development)	Stratify by domain, difficulty, query type; maintain proportional representation
D: Version Control & Storage	S3 (eu-central-1) with Object Lock; Phoenix Datasets for experiment management; DVC for version lineage; JSONL schema with full validation	Test/Golden split locked immediately in S3 Object Lock Compliance mode — no post-hoc modification

6.3 Anthropic/Claude Benchmarking Pattern — Replicated

Axis	Method	Implementation
Capability mapping	Map every test case to a specific capability: reasoning, coding, maths, world knowledge, safety, instruction following	Custom Case taxonomy with capability_tag metadata field
Multi-axis scoring	Human eval 1-7 scale AND auto eval 0-1 AND safety eval pass/fail — all three required simultaneously for high-risk decisions	Strands Evals Experiment with HelpfulnessEvaluator + OutputEvaluator + HarmfulnessEvaluator combined
Red-team testing	Adversarial probing per safety category; jailbreak resistance; instruction hierarchy violations; constitutional AI alignment	Custom adversarial Case set (minimum 50 cases per risk category)
ELO comparison	Side-by-side blind evaluation; diverse evaluator pool; model identity never revealed to human judges	Phoenix Experiments A/B comparison; randomise response order to eliminate position bias
Regression suite	Every fixed failure becomes a permanent test — zero regression policy strictly enforced for all future deployments	Strands Evals CI/CD gate with report.assert_pass() blocking merge on any regression
Statistical validation	Bootstrap confidence intervals; $p < 0.05$ threshold for any claimed improvement; minimum 200 samples per capability axis	Python scipy for significance tests; DeepEval statistical utilities

7.1 Alert Thresholds & Escalation Matrix

Metric	Warning	Critical	Auto-Action	Human Escalation
Helpfulness score	< 0.75	< 0.60	Increase sampling to 50%; trigger on-demand eval	AI Risk team + on-call SRE
Tool Selection Accuracy	< 0.85	< 0.70	Trigger on-demand eval batch on recent traces	Platform engineering
Faithfulness (RAG)	< 0.80	< 0.65	Flag knowledge base for review; increase sampling	Knowledge management team
Task Success Rate	< 0.70	< 0.55	Enable Level 2 HITL for all sessions	Product team + CRO
Input embedding drift	> 0.10	> 0.20	Increase sampling; alert team; begin investigation	ML Engineering
Output embedding drift	> 0.15	> 0.25	Pause high-risk automated decisions	AI Ethics Committee
PII Leakage rate	> 0.001	> 0.01	EMERGENCY STOP — block all outputs immediately	DPO + CISO + Legal (72h GDPR clock starts)
Latency P95 (ms)	> 3000	> 5000	Scale-out trigger; check downstream tools	Infrastructure / SRE
Error rate (%)	> 1%	> 5%	Traffic rerouting; prepare rollback	On-call SRE + CTO
AML False Positive rate	> 0.35	> 0.50	Pause AML auto-flagging; switch to manual review	MLRO + Compliance officer

7.2 Automated Remediation Pipeline

Drift detection triggers a Lambda-based auto-remediation chain: CloudWatch Alarm -> SNS topic -> Lambda function. The Lambda performs: (1) increase sampling rate; (2) trigger on-demand evaluation on recent failing traces; (3) create incident ticket in Jira; (4) notify on-call via PagerDuty; (5) post summary to Slack. If the incident meets the DORA 'significant' threshold, a separate Lambda notifies the regulator within the required 4-hour window.

CHAPTER 8 — EU BANKING COMPLIANCE FRAMEWORK

8.1 Regulatory Landscape

Regulation	In Force	Key AI Obligations	Max Penalty
EU AI Act (2024/1689)	Aug 2024 (phased to 2027)	High-risk conformity Art.10-15; human oversight; audit trail 10yr; EU AI database registration	EUR 35M or 7% global turnover
GDPR (2016/679)	May 2018 (active)	PII minimisation; right to explanation (Art.22); right to erasure; DPIA for high-risk AI; data transfer restrictions	EUR 20M or 4% global turnover
DORA (2022/2554)	January 2025 (active)	ICT risk management; incident reporting within 4h; annual resilience testing; third-party ICT risk (AWS, Anthropic, Arize)	Up to EUR 5M per entity
AMLD6 (2018/1673)	Active	AML/CTF controls; KYC verification; SAR justification; FATF compliance for AI-generated flags; demographic non-discrimination	Criminal liability + fines
CRR III / CRD VI	2025	Credit model validation; IRB model alignment for AI credit scoring; capital requirements for AI-related risk exposures	Capital add-on requirements
EBA Guidelines	Ongoing	Loan origination and monitoring; model risk management; internal governance; PD/LGD estimation guidelines for AI models	Supervisory action; removal of model approval

8.2 High-Risk AI Classification — Banking Use Cases

Risk Level	Use Case	Full Obligations Aug 2026?	Key Evaluation Gate
HIGH-RISK (Annex III)	Credit scoring and creditworthiness assessment of natural persons	YES — Art.10-15 + conformity assessment + EU DB registration	SHAP explanation + exact trajectory + HITL human review check
HIGH-RISK (Annex III)	Life and health insurance risk assessment and pricing for individuals	YES — Art.10-15 + conformity	Demographic parity + fairness rubric + plain-language explanation
HIGH-RISK (Annex III)	AML and KYC risk profiling of natural persons	YES — Art.10-15 + conformity	SAR quality rubric + demographic bias check + audit trail 100%
HIGH-RISK (Annex III)	Eligibility assessment for essential financial services	YES — Art.10-15 + conformity	Disparate impact ratio (DIR >= 0.80) + adverse action notice quality
LIMITED-RISK (Art.52)	Customer service AI chatbot or virtual banking assistant	Transparency only: disclose AI interaction to user	AI disclosure rate = 100%, HarmfulnessEvaluator on every session
LIMITED-RISK (Art.52)	AI-generated financial content, summaries, or explanations	Label AI-generated content as such	Faithfulness + Groundedness + content labelling compliance check
MINIMAL-RISK	Internal operations automation, document processing, reporting	Voluntary codes of conduct only	Standard quality and safety metrics from Chapter 5
PROHIBITED (since Feb 2025)	Social scoring of EU citizens; subliminal manipulation of financial decisions	BANNED — do not build these systems	Block at system design; never deploy; audit any legacy systems

8.3 PII Detection — Three-Layer Architecture

Layer	Tools	Applied When	Action on Detection	EU Requirement
Layer 1: Pre-LLM input scrubbing	AWS Comprehend (entity detection) + Microsoft Presidio (EU-specific: IBAN, VAT, NHI)	Before every LLM call — synchronous blocking check	REDACT entities with [REDACTED_TYPE]; PSEUDONYMISE with consistent token; BLOCK if Art.9 special category data	GDPR Art.5(1)(c) data minimisation; Art.25 privacy by design and by default

Layer	Tools	Applied When	Action on Detection	EU Requirement
Layer 2: Output guard	AgentCore Builtin.PIILeakage evaluator + Phoenix custom LLM-Judge evaluator	After every agent response — 100% sampling, never reduce	BLOCK output; log incident with PII category and session ID; notify DPO immediately; create GDPR breach assessment	GDPR Art.5 security; EU AI Act Art.10 data governance for high-risk systems
Layer 3: Storage anonymisation	AWS Macie (S3 PII scanning) + CloudWatch Logs Insights PII pattern detection	Before storing any traces, logs, or evaluation data anywhere	Pseudonymise all customer IDs; use synthetic PII in golden datasets (never real customer data); enforce S3 bucket policies	GDPR Art.6 lawful basis documentation; Art.44-46 data transfer restrictions to non-EU

8.4 Fairness & Bias Evaluation

Metric	Formula	Target	Tool	Regulatory Basis
Disparate Impact Ratio (DIR)	$P(\text{approve} \mid \text{minority group})$ divided by $P(\text{approve} \mid \text{majority group})$	≥ 0.80 (EU 4/5ths rule)	Custom Strands Cases + batch scoring on balanced test set	EU AI Act Art.10; EU anti-discrimination law
Demographic Parity Difference	Absolute value of $P(y_{\text{hat}}=1 \mid \text{GroupA})$ minus $P(y_{\text{hat}}=1 \mid \text{GroupB})$	≤ 0.05	Custom evaluator on balanced demographic test set	EU AI Act Art.10 non-discriminatory training data
Equal Opportunity Gap	Absolute value of $\text{TPR}_{\text{GroupA}}$ minus $\text{TPR}_{\text{GroupB}}$	≤ 0.05	Recall calculation stratified by protected demographic	EU AI Act Art.10; EBA fair lending guidelines
Counterfactual Fairness	Cosine distance between outputs for same query with only protected attribute changed (name, gender, nationality)	≤ 0.15	50 name-swap Cases — identical query, swapped protected attribute	EU AI Act Art.10; GDPR Art.22 individual decision fairness
Calibration across groups	$P(y=1 \mid \text{score}=p, \text{GroupA})$ approximately equals $P(y=1 \mid \text{score}=p, \text{GroupB})$	Calibration error ≤ 0.05	Reliability diagram per demographic group on test set	EBA credit model governance; CRR III model validation

8.5 Art. 12 Audit Log Schema — Mandatory Fields

Every HIGH-RISK agent interaction must produce an immutable audit log. Storage: S3 Object Lock (Compliance mode), eu-central-1 region, AES-256 encryption, 10-year retention minimum.

Field	Type	Content	Regulatory Purpose
log_id	UUID v4	Globally unique per interaction	Traceability — enables point-in-time audit lookup
timestamp_utc	ISO-8601	Millisecond-precision UTC timestamp	Audit sequencing and incident timeline reconstruction
session_id	Pseudonymised	Customer session ID — no raw PII	GDPR Art.5(1)(c) data minimisation in logs
agent_id + version	String	Agent ARN + semantic version tag (e.g. v2.1.0)	Version accountability — which code made this decision
use_case_classification	Enum	HIGH_RISK or LIMITED or MINIMAL	EU AI Act Art.11 classification evidence
input (redacted)	String	Query with PII replaced by type tokens [IBAN], [NAME]	GDPR Art.5 minimisation; preserves audit without PII
pii_detected	Array	PII categories found + action taken (REDACT/BLOCK)	GDPR compliance evidence; breach assessment support
reasoning_trace	Array	All LLM calls + tool calls with parameters and outputs	EU AI Act Art.12 automatic logging of events
output	Encrypted ref	Response stored separately, AES-256, EU region only	GDPR Art.32 security of processing

Field	Type	Content	Regulatory Purpose
evaluation_scores	Object	All metric scores for this interaction	Quality and compliance evidence per interaction
human_review	Object	Required flag + reviewer ID + decision + review timestamp	EU AI Act Art.14 human oversight evidence
gdpr_basis	String	Lawful basis: contract / consent / legal obligation	GDPR Art.6 lawful basis documentation
retention_end_date	ISO date	Creation date plus 10 years for HIGH_RISK systems	EU AI Act Art.12 minimum retention period
data_residency	AWS Region ID	Must be EU region (eu-central-1, eu-west-1, etc.)	GDPR Art.44-46 data transfer restrictions

CHAPTER 9 — COMPLETE LIFECYCLE JOURNEY (IDEATION TO RETIREMENT)

PHASE 1

IDEATION & REQUIREMENT CAPTURE

- Define agent purpose, user personas, scope, and measurable success criteria
- Classify use case: HIGH_RISK / LIMITED / MINIMAL under EU AI Act Annex III (mandatory first step)
- Map to evaluation planes: which of Response / RAG / Agentic metrics apply to this agent?
- Select AgentCore built-in evaluators + identify gaps requiring custom evaluators
- Define acceptance thresholds per metric — document formally in AI Governance Policy
- Choose judge model: Claude Sonnet (primary) + Nova Premier (secondary for ensemble judging)
- Complete DPIA if HIGH_RISK — obtain DPO sign-off before any customer data processing
- Register planned AI system with DPO and AI Risk function — begin technical documentation draft

PHASE 2

BENCHMARK & DATASET CONSTRUCTION

- Run ExperimentGenerator with agent context + task description — generate 50-100 seed cases
- SME review and curation: remove duplicates, adjust difficulty distribution (20% hard minimum)
- Red-team sessions: generate adversarial, edge case, jailbreak, and safety test cases
- Human annotation: ground truth + rubrics targeting inter-annotator kappa ≥ 0.70
- Create counterfactual fairness suite: 50 name / gender / nationality swap pairs
- Create PII leakage test suite: 100 cases with realistic synthetic PII in inputs and context
- Create EU compliance suite: adverse action notices, AML SAR quality, AI disclosure tests
- Version-control in S3 (eu-central-1) + Phoenix Datasets + DVC — lock golden split immediately

PHASE 3

BUILD & INSTRUMENT

- Implement Strands Agent with tools, system prompt, and OTEL instrumentation via OpenInference
- Register OpenInference spans -> Phoenix project and AgentCore OTEL collector simultaneously
- Configure AgentCore Online Evaluation: start at 10% sampling with initial evaluator set
- Implement PII Layer 1 guard (Presidio + AWS Comprehend) as pre-processor before every LLM call
- Build custom evaluators: PII output guard, EU fairness rubric, GDPR Art.22 adverse action checker
- Configure AgentCore Policy with Cedar rules: tool access, session isolation, output constraints
- Set up HITL workflow: Phoenix annotation queue + Jira + reviewer SLA (4h credit, 48h AML)
- Configure Langfuse for development trace collection and per-session cost tracking

PHASE 4

PRE-DEPLOYMENT EVALUATION GATE (CI/CD)

- Gate 1 — Prohibited practices: automated scan, HARD BLOCK on any positive finding
- Gate 2 — PII safety: 100 synthetic PII cases, zero tolerance, HARD BLOCK on any single leak
- Gate 3 — Fairness: 50 counterfactual pairs, DIR ≥ 0.80 , sentiment delta ≤ 0.05 required
- Gate 4 — Quality: full 200-500 case benchmark, all quality thresholds must pass simultaneously
- Gate 5 — Trajectory compliance: EU-mandated tool sequences verified with exact match scoring
- Gate 6 — Explainability: 100% explanation coverage for all HIGH_RISK decision outputs
- Gate 7 — Audit trail: 100% of test interactions produce valid Art.12 log entries with all fields
- Gate 8 — Safety: HarmfulnessEvaluator + RefusalQuality on full adversarial test set
- Use `report.assert_pass(threshold=N)` — merge is blocked automatically if any single gate fails
- Require: MRM independent validation sign-off + DPO review before any production deployment

PHASE 5**DEPLOY WITH ONLINE MONITORING**

- Enable AgentCore Online Evaluation: start 10% sampling across all configured evaluators
- Activate Phoenix drift monitors: input embedding, output embedding, metric EWMA, feature drift
- HITL routing active: all HIGH_RISK decisions automatically routed to Level 2 review queue
- Enable 100% sampling for HarmfulnessEvaluator and PIILeakage — never sample safety checks
- CloudWatch alarms active: quality thresholds, latency P95, error rate, cost per session
- Daily automated compliance email to AI Risk function, DPO, and CISO
- Notify regulators if required: EU AI database registration + any required pre-notifications

PHASE 6**CONTINUOUS IMPROVEMENT LOOP**

- Weekly: pull failing traces from Phoenix; run cluster heatmap; identify top-5 failure patterns
- Weekly: review HITL decisions — learn from human reviewer modifications and rejections
- Monthly: full benchmark regression suite to detect gradual quality decay before users notice
- Monthly: fairness audit — compute DIR and demographic parity across all protected attributes
- Monthly: auto-generate compliance report (PII events, fairness metrics, HITL stats, drift events)
- Curate failure cases into golden dataset — augment with newly discovered edge cases
- Prompt engineering iteration -> Strands Evals experiment -> comparison vs production baseline
- Re-run full gate suite before merging any improvement; track trend across Phoenix Experiments

PHASE 7**DRIFT DETECTION & INCIDENT RESPONSE**

- Drift detected by Phoenix monitor or CloudWatch alarm -> SNS -> Lambda auto-response
- Lambda: increase sampling 10%->50%; trigger on-demand eval; create incident ticket in Jira
- Triage (0-15 min): scope check; deployment correlation; time-of-day pattern analysis
- Diagnose (15-60 min): Phoenix cluster analysis; failure mode classification (data/concept/model/performance)
- DORA: if 'significant' threshold met -> notify regulator within 4 hours of classification
- Remediate: rollback OR prompt fix OR tool API fix OR knowledge base update -> re-evaluate
- Validate: re-run full benchmark; confirm all metrics above thresholds; re-enable full traffic
- Post-mortem: document root cause; update runbook; add failure case to regression suite

PHASE 8**MODEL UPGRADE EVALUATION**

- Run full benchmark on new model version using offline task function on same golden dataset
- Statistical significance: $n \geq 200$ samples, $p < 0.05$ threshold for any claimed improvement
- Zero regression policy: all existing passing cases must continue to pass on new model version
- EU compliance re-check: fairness suite + PII suite + trajectory compliance on new model
- Cost and latency analysis: new model within +20% cost and +10% P95 latency of current baseline
- Canary deployment: 5% of traffic to new model, monitor for 48 hours before proceeding
- Progressive rollout: 5% -> 25% -> 50% -> 100% with full evaluation gate at each stage
- Update benchmark baseline to new model scores; update Art.11 technical documentation

PHASE 9**RETIREMENT & ARCHIVAL**

- Sunset criteria: metric floor breached persistently (3 consecutive monthly reviews) OR successor validated
- Retirement decision: AI Governance Committee approval required; Board approval for HIGH_RISK
- Archive all traces: S3 Object Lock Compliance mode; retain minimum 10 years (EU AI Act Art.12)
- Archive: all evaluation reports, benchmark results, DPIA, conformity assessment, incident log
- Generate final performance audit report: lifecycle evaluation trend, incident history, drift events
- Baseline comparison document: how successor compares across every metric dimension
- De-register from EU AI database (if applicable); notify DPO and relevant national supervisors
- GDPR Art.17 erasure: delete all non-archived personal data; document deletion for DPA evidence
- Knowledge transfer: share lessons learned with teams building and operating successor systems

CHAPTER 10 — INTEGRATION STACK & QUICK REFERENCE

10.1 Tool Comparison Matrix

Capability	Strands Evals	AgentCore Evals	Arize Phoenix	RAGAS	DeepEval
Online (production) eval	No	Yes — native	Yes — online evals	No	No
On-demand / CI/CD eval	Yes — native	Yes — native	Yes — experiments	Yes	Yes
RAG quality metrics	Yes — custom	Yes — built-in	Yes — templates	Yes — best	Yes
Agentic / tool metrics	Yes — 4 built-in	Yes — 3 built-in	Yes — custom	Yes — Agent Goal	Yes — DAG
Multi-turn simulation	Yes — ActorSimulator	No	No	No	Yes — conversational
Drift detection	No	Yes — via CloudWatch	Yes — native (best)	No	No
Human annotation	No	No	Yes — annotation queue	No	Yes — via Confident AI
EU self-hostable	Yes (AWS EU)	Yes (AWS EU regions)	Yes — EKS Frankfurt	Yes — self-hosted	Yes — self-hosted
Strands native integration	Yes	Yes — via OTEL	Yes — via OTEL	Yes — via Langfuse	Yes — via API
Open source	Yes	No — AWS managed	Yes (7800+ stars)	Yes	Yes (400k+ downloads)

10.2 Complete Metric Threshold Reference Card

Metric	Target	Warning	Critical	Zero Tolerance	Regulation
Helpfulness	≥ 0.80	< 0.75	< 0.60	No	EBA consumer outcomes
Correctness	≥ 0.82	< 0.70	< 0.55	No	EU AI Act Art.15
Groundedness / Faithfulness	≥ 0.85	< 0.75	< 0.65	No	GDPR Art.5 accuracy
Tool Selection Accuracy	≥ 0.90	< 0.85	< 0.70	No	EU AI Act Art.15
Tool Parameter Accuracy	≥ 0.88	< 0.80	< 0.68	No	EU AI Act Art.15
Task / Goal Success Rate	≥ 0.75	< 0.70	< 0.55	No	EBA consumer outcomes
Harmlessness	$= 1.00$	< 0.99	< 0.95	YES — any fail	EU AI Act prohibited
PII Leakage Rate	$= 0.000$	any > 0	any > 0	YES — any leak = block	GDPR Art.5 and Art.25
AI Disclosure Rate	$= 1.000$	< 1.000	< 1.000	YES	EU AI Act Art.52
Audit Trail Coverage	$= 1.000$	< 1.000	< 0.990	YES for high-risk	EU AI Act Art.12
Disparate Impact Ratio	≥ 0.80	< 0.85	< 0.80	YES — block deploy	EU AI Act Art.10
Demographic Parity Diff.	≤ 0.05	> 0.04	> 0.05	YES — block deploy	EU AI Act Art.10
SAR Explanation Quality	≥ 0.90	< 0.88	< 0.80	No	AMLD6, FATF
Adverse Action Quality	≥ 0.95	< 0.93	< 0.85	No	GDPR Art.22

Metric	Target	Warning	Critical	Zero Tolerance	Regulation
Input Embedding Drift	<= 0.08	> 0.10	> 0.20	No	EU AI Act Art.9
Agent Uptime	>= 99.5%	< 99.5%	< 99.0%	No	DORA ICT availability
MTTD — incident detection	<= 15 min	> 20 min	> 30 min	No	DORA incident management
MTTR — incident response	<= 4 hours	> 4 hours	> 8 hours	No	DORA incident management
Latency P95 (ms)	<= 2500	> 3000	> 5000	No	DORA resilience
Error Rate (%)	<= 0.5%	> 1.0%	> 5.0%	No	DORA incident classification

10.3 Regulatory Compliance Calendar

Deadline	Regulation	Action Required	Status
Feb 2025 — DONE	EU AI Act	Cease all 8 categories of prohibited AI practices; begin AI literacy programme for all staff	Verify complete — if not, URGENT action required now
Jan 2025 — DONE	DORA	Full ICT risk framework live; incident detection and reporting operational; third-party register	Verify complete — if not, URGENT action required now
Aug 2025 — DONE	EU AI Act	GPAI model governance obligations; AI system inventory completed and classified	Verify complete — if not, URGENT action required now
NOW (Q2 2026)	GDPR	DPIAs completed for all HIGH_RISK AI systems; DPAs signed with AWS, Anthropic, Arize	ACTION REQUIRED — DPO must sign off before any deployment
NOW (Q2 2026)	EU AI Act	Technical documentation (Art.11) drafted; conformity assessment started for HIGH_RISK	ACTION REQUIRED — engage MRM and Legal immediately
NOW (Q2 2026)	All	Deploy this evaluation framework; all 8 CI/CD gates operational; Phoenix monitoring live	THIS DOCUMENT — implementation timeline required
Aug 2026	EU AI Act	HIGH_RISK AI: full conformity + CE marking + EU AI database registration complete	CRITICAL DEADLINE — 4 months away — escalate to board
Aug 2026	EU AI Act	Art.12 audit trail + Art.14 human oversight + Art.13 transparency mechanisms all live	CRITICAL DEADLINE — must be live before this date
Aug 2027	EU AI Act	All remaining provisions; national market surveillance authority registration	Future planning — begin roadmap now
Dec 2030	EU AI Act	Large-scale IT systems extended transition period ends — all IT-embedded AI must comply	Long-term enterprise architecture roadmap
Quarterly (recurring)	EBA	Independent model validation report; ongoing monitoring review by MRM team	Recurring — schedule in AI Governance Calendar
Annually (recurring)	DORA	Full ICT risk assessment review; resilience testing (TLPT every 3 years for significant)	Recurring — coordinate with CISO and Operations

10.4 Escalation Playbooks

Alert Scenario	Immediate Action (0-15 min)	Investigation (15-60 min)	Escalation Path
PII leakage detected in output	BLOCK all outputs; emergency stop agent; DPO notification required within 1 hour of discovery	Identify affected sessions; scope of leak; assess whether GDPR 72h notification threshold is met	DPO + CISO + Legal; potential GDPR Art.33 breach notification to national DPA within 72h
Quality score collapse (>20% drop)	Increase sampling to 50%; check deployment log for any recent changes in past 24h	Phoenix cluster analysis; identify failure mode: data drift vs model change vs tool API issue	AI Risk team + on-call SRE; rollback if correlated with recent deployment
Fairness / DIR breach below 0.80	Pause all HIGH_RISK automated decisions immediately; route 100% to Level 3 HITL	Compute DIR across all demographic groups; identify which group is affected and by how much	AI Ethics Committee + CRO + Legal; potential regulator notification (EBA / national CA)
AML false positive surge >35%	Pause AML auto-flagging; switch entirely to manual review queue; notify MLRO	Analyse flagged transaction characteristics; identify model or data change driving surge	MLRO + Compliance + Risk; assess AMLD6 obligations; potential FATF notification
DORA significant ICT incident	Incident ticket created; war-room assembled; begin impact assessment for 'significant' classification	Containment actions executed; RCA commenced; business continuity plan activated if needed	CIO + CISO; if 'significant': mandatory regulator notification within 4h of classification decision
Embedding drift score >0.20	Auto-increase sampling; trigger on-demand eval batch on traces from past 6 hours	Determine if input query distribution or model output has semantically shifted; compare topic distributions	ML Engineering; prompt or model update cycle; MRM review if change is significant for HIGH_RISK models

DOCUMENT INFORMATION & GOVERNANCE

Attribute	Value
Document Title	Complete AI Agent Evaluation Framework — EU Banking Edition v2.0
Framework Stack	AWS Bedrock AgentCore + Strands Evals (10 evaluators) + Arize Phoenix + RAGAS + DeepEval
Regulatory Coverage	EU AI Act (2024/1689) + GDPR (2016/679) + DORA (2022/2554) + AMLD6 + CRR III + EBA Guidelines
Source Material	AWS ML Blog: Strands Evals guide (Mar 18 2026); AgentCore docs; EBA AI Act factsheet (Nov 2025); EDPB LLM Privacy Risk report (Apr 2025)
Chapters	10 chapters covering: architecture, Strands Evals, AgentCore, Phoenix, metrics, benchmarks, drift, EU compliance, lifecycle, quick reference
Metrics Defined	45+ metrics across 4 planes: response quality, RAG quality, agentic behaviour, operational/infrastructure, and EU compliance
Lifecycle Phases	9 phases: Ideation, Benchmark, Build, Pre-Deploy Gate, Deploy, Improve, Drift Response, Model Upgrade, Retirement
Critical Deadline	August 2, 2026 — EU AI Act HIGH-RISK AI full obligations come into force — all conformity must be complete before this date
Classification	Confidential — Internal Use — Do Not Distribute Externally Without Legal Approval
Owner	AI Platform Engineering Team
Review Cycle	Quarterly — or immediately upon any EU regulatory update or significant incident
Generated	06 April 2026

Sources: AWS Bedrock AgentCore documentation; AWS Machine Learning Blog — 'Evaluating AI agents for production: A practical guide to Strands Evals' (March 18, 2026); EBA AI Act implications factsheet (November 2025); EU AI Act text (Regulation 2024/1689); GDPR (Regulation 2016/679); DORA (Regulation 2022/2554); European Data Protection Board LLM Privacy Risk report (April 2025); Arize Phoenix documentation; RAGAS framework documentation; DeepEval documentation; CFA Institute Explainable AI in Finance report (2025); IOSCO AI in Capital Markets consultation (March 2025).